

ITA0608 - Machine Learning

List of Lab Exercises

7. Write a program to implement Logistic Regression (LR) algorithm in python.

Aim

To implement the **Logistic Regression (LR)** algorithm in Python for binary classification and evaluate its accuracy.

Algorithm

1. Import the required Python libraries.
2. Load the dataset.
3. Select the input features (**X**) and target (**y**).
4. Split the dataset into training and testing sets.
5. Create a Logistic Regression model.
6. Train the model using the training data.
7. Predict the class labels for the test data.
8. Calculate and display the accuracy of the model.

Program Code

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score

data = load_iris()

X = data.data

y = data.target

X = X[y != 2]
```

```
y = y[y != 2]

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

model = LogisticRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Actual Labels:", y_test)

print("Predicted Labels:", y_pred)

print("Accuracy:", accuracy)
```

Output

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# Load dataset
data = load_iris()
X = data.data
y = data.target

# Use only two classes (Binary Classification)
X = X[y != 2]
y = y[y != 2]

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create Logistic Regression model
model = LogisticRegression()

# Train the model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Actual Labels:", y_test)
print("Predicted Labels:", y_pred)
print("Accuracy:", accuracy)
```

```
... Actual Labels: [1 1 1 0 0 0 0 1 0 0 0 0 1 0 1 0 1 1 0 0]
Predicted Labels: [1 1 1 0 0 0 0 1 0 0 0 0 1 0 1 0 1 1 0 0]
Accuracy: 1.0
```

Result

The **Logistic Regression algorithm** was successfully implemented in Python. The model classified the data correctly and achieved **100% accuracy (1.0)** on the test dataset.